

AI335L Deep Learning Lab
Experiment No. 10 — Phase 3

Skin Cancer Detection

Using Deep Learning

Phase 3 Architecture Report

CNN Track · SkinCancerCNN with BatchNorm, Dropout & Residual Block

Course	Deep Learning Lab — AI335L
Project Title	Skin Cancer Detection Using Deep Learning
Architecture Track	CNN Track
Deep Model	SkinCancerCNN with BatchNorm, Dropout & Residual Block
Phase 2 Baseline	MLP — Test Accuracy: 67.88%, Test Loss: 0.6199
Team Members	Hafsa Ashfaq, zain bin ibrar Maryam Shahid
GitHub Repository	https://github.com/hafsa-khan12/DermaVision-AI-Skin-Cancer-Diagnosis/tree/main/Phase-4/reports
Commit Hash / Tag	phase3-submission

1. Introduction

Phase 3 moves the project from a simple Phase 2 MLP baseline toward a deep architecture that is more appropriate for image data. The current project focuses on binary skin cancer classification using skin lesion images, classifying each input image as either benign or malignant.

The Phase 2 MLP provided an honest baseline accuracy of 67.88%. However, the MLP flattened each image into a one-dimensional vector, losing spatial information such as lesion border, texture, and color arrangement. Phase 3 therefore introduces a convolutional neural network because CNNs preserve local spatial relationships and learn image-specific features through filters.

The main objective of Phase 3 is to implement, train, and evaluate a CNN model using the same dataset pipeline as Phase 2, then compare its performance with the MLP baseline under fair conditions.

2. Architecture Justification

2.1 Selected Track: CNN

The selected architecture track is CNN because the dataset contains 2D skin lesion images. CNNs are designed for visual data and can learn local patterns such as lesion shape, texture, color variation, and edge irregularity — properties essential for distinguishing benign and malignant skin lesions.

2.2 Specific Architecture Variant

The implemented model is named SkinCancerCNN: a custom CNN containing four convolution blocks, BatchNorm layers, MaxPooling, one residual block, global average pooling, dropout, and a two-layer classifier head. The model is intentionally smaller than large pretrained models to be trainable on limited hardware.

2.3 Training Strategy

The model is trained from scratch rather than using a pretrained backbone in this phase. This allows the project to demonstrate understanding of convolution, pooling, normalization, regularization, and residual connections before moving to transfer learning in later phases.

2.4 Input and Output Shape

Component	Shape / Description
Input image	Batch × 3 × 224 × 224
Output logits	Batch × 2
Classes	Benign, Malignant
Approx. trainable parameters	718,210

2.5 Inductive Bias

CNNs have useful inductive biases for image problems. Local connectivity allows the model to learn small visual structures, weight sharing reduces parameter count compared with fully connected networks, and pooling adds partial translation invariance — properties that match the skin lesion classification problem.

3. Model Architecture

The model architecture progressively reduces spatial resolution while increasing channel depth, allowing the network to learn low-level features in early layers and more abstract lesion patterns in deeper layers.

[Figure 1: SkinCancerCNN Architecture Diagram]

3.1 Layer-by-Layer Architecture

Layer / Block	Operation	Output Shape	Purpose
Input	RGB image resized to 224×224	[B, 3, 224, 224]	Standardized input
Conv Block 1	Conv2D 3→32, BatchNorm, ReLU, MaxPool	[B, 32, 112, 112]	Low-level edges and gradients
Conv Block 2	Conv2D 32→64, BatchNorm, ReLU, MaxPool	[B, 64, 56, 56]	Medium-level lesion patterns
Conv Block 3	Conv2D 64→128, BatchNorm, ReLU, MaxPool	[B, 128, 28, 28]	Deeper spatial features
Residual Block	Two Conv2D layers with skip connection	[B, 128, 28, 28]	Gradient flow and feature reuse
Conv Block 4	Conv2D 128→256, BatchNorm, ReLU, MaxPool	[B, 256, 14, 14]	High-level discriminative features
Global Avg Pool	AdaptiveAvgPool2D	[B, 256, 1, 1]	Reduces spatial dims without flattening
Classifier	Dropout → Linear 256→128 → ReLU → Dropout → Linear 128→2	[B, 2]	Benign/malignant logits

3.2 Receptive Field Discussion

The final convolutional feature map has a spatial size of 14×14 after four pooling operations. The receptive field grows through repeated 3×3 convolution and pooling layers, allowing deeper activations to summarize increasingly larger regions of the skin lesion image. The use of global average pooling ensures that all final feature map locations contribute to the classifier decision.

4. Training Setup

The training pipeline was upgraded from Phase 2 to support config-driven training, scheduler-based learning rate updates, checkpoint saving, gradient norm logging, and gradient clipping. The same dataset pipeline and preprocessing decisions as the Phase 2 baseline are used to keep comparison fair.

Hyperparameter	Value
Image size	224 × 224
Batch size	32
Epochs	10
Optimizer	Adam
Learning rate	0.0003
Weight decay	0.0001
Loss function	CrossEntropyLoss
Scheduler	StepLR — step size 3, gamma 0.5
Dropout	0.3
Gradient clipping	Max norm 1.0
Seed	42
Checkpoint	experiments/checkpoints/cnn_v1_best.pth
Log file	experiments/logs/cnn_v1_log.csv

[Figure 2: CNN Training Loss and Validation Loss Curves (cnn_loss_curve.png)]

[Figure 3: CNN Accuracy Curve (cnn_accuracy_curve.png)]

[Figure 4: CNN Gradient Norm Curve (cnn_gradient_norm.png)]

5. Regularization Ablation

Phase 3 requires three configurations: unregularized, regularized, and normalized. The purpose of this ablation is to observe how dropout, weight decay, augmentation, and BatchNorm affect overfitting and validation performance.

Configuration	Model Settings	Expected Observation
Unregularized	No dropout, no weight decay, no augmentation, BatchNorm disabled	Model may memorize training data and show overfitting
Regularized	Dropout 0.3, weight decay 0.0001, augmentation enabled	Reduced overfitting and improved validation stability

Configuration	Model Settings	Expected Observation
Normalized	Regularized setup plus BatchNorm	More stable gradients and smoother training

[Figure 5: Ablation Curves — Train Loss, Validation Loss, and Accuracy for All Three Runs]

6. Comparison with Phase 2 Baseline

The Phase 2 baseline was a feedforward MLP. The Phase 3 model is a CNN. Both models use the same benign/malignant dataset and the same input image size. The comparison is fair because both models are trained using the same data loading and preprocessing pipeline.

Model	Architecture Type	Parameters	Test Loss	Test Accuracy	Notes
Phase 2 Baseline MLP	Fully connected NN	[Add]	0.6199	67.88%	Flattens images, loses spatial structure
Phase 3 SkinCancerCNN	Convolutional NN	718,210	[Add CNN test loss]	[Add CNN test accuracy]	Preserves spatial structure, learns lesion features

7. Interpretation and Visualization

For CNN interpretation, the recommended visualizations are feature maps, saliency maps, or Grad-CAM. Grad-CAM is especially useful for medical image projects because it highlights the image regions that contributed most strongly to the prediction. A useful Grad-CAM result should highlight the lesion area rather than irrelevant background.

[Figure 6: CNN Confusion Matrix (cnn_confusion_matrix.png)]

[Figure 7: Grad-CAM or Feature Map Visualization]

8. Failure Analysis

Failure analysis examines 5–10 images where the model made incorrect predictions, helping identify what types of cases are difficult and what improvements are required in later phases.

Failure Pattern	Possible Reason	Planned Improvement
Malignant predicted as benign	Subtle lesion features or low contrast	Stronger augmentation, class weighting, or transfer learning

Failure Pattern	Possible Reason	Planned Improvement
Benign predicted as malignant	Benign lesion visually similar to malignant examples	More diverse benign samples and false-positive analysis
Wrong prediction on dark/bright images	Lighting variation affects features	Improved normalization and brightness augmentation
Heatmap focuses on background	Model may learn artifacts or dataset bias	Cropping, segmentation, or better preprocessing

[Figure 8: Wrong Prediction Examples (5–10 samples with predicted class and confidence)]

9. Issues Encountered & Plan for Phase 4

During Phase 3, the major issues were project structure setup, missing config file location, module import errors, and ensuring training logs and checkpoints were saved correctly. These were fixed by moving the config file to the project root, adding `__init__.py` files, and using consistent paths.

The plan for Phase 4 is to improve performance using transfer learning and hyperparameter tuning. A pretrained backbone such as ResNet-18 can be used because it has already learned strong visual features from large-scale image datasets.

Phase 4 Task	Planned Action
Transfer learning	Train a pretrained ResNet-18 or EfficientNet model using transfer learning
Baseline comparison	Compare pretrained CNN results with custom CNN and Phase 2 MLP
Hyperparameter tuning	Tune learning rate, dropout, weight decay, and augmentation strength

10. Reproducibility and Submission Checklist

Artifact	Location / Status
GitHub repository	https://github.com/hafsa-khan12/DermaVision-AI-Skin-Cancer-Diagnosis/tree/main/Phase-4/reports
Commit hash / tag	phase3-submission
Model code	src/models/cnn.py
Training script	src/training/train_cnn.py
Evaluation script	src/evaluation/evaluate_cnn.py
Config file	config.yaml (project root)
Training log	experiments/logs/cnn_v1_log.csv
Best checkpoint	experiments/checkpoints/cnn_v1_best.pth

Artifact	Location / Status
Figures	reports/figures/
This report	reports/phase3_architecture_report.docx